

Absolute-Sum Stability for the Real Lipschitz Numerical Index Question

Automatic math-discovery packet

June 13, 2026

Source question

The source paper is Antonio Perez-Hernandez, *Lipschitz vs Linear Numerical Index in certain Banach spaces*, arXiv:2507.18208. It studies the question, originally posed by Wang–Huang–Tan, whether the Lipschitz numerical index $n_L(X)$ and the classical numerical index $n(X)$ always coincide.

$$\max_{|\theta|=1} \|\text{Id}_X + \theta T\| \geq 1 + \alpha \|T\|$$

for every operator T , see e.g. [17, Proposition 3.3]. In particular, if $n(X) > 0$ then Id_X is a strongly extreme point of the unit ball of $\mathcal{L}(X)$.

These notions have been extended in several directions. In this work, we focus on a non-linear variant introduced by X. Huang, D. Tan, and R. Wang in [20], known as the *Lipschitz numerical index*. Let $\text{Lip}_0(X)$ denote the Banach space of all Lipschitz maps $F : X \rightarrow X$ such that $F(0) = 0$, equipped with the norm

$$\|F\|_L := \sup \left\{ \frac{\|F(x_1) - F(x_2)\|}{\|x_1 - x_2\|} : x_1, x_2 \in X, x_1 \neq x_2 \right\} < \infty.$$

For every $F \in \text{Lip}_0(X)$, the *numerical range* of F is defined by

$$W_L(F) = \left\{ \frac{\langle x^*, F(x_1) - F(x_2) \rangle}{\|x_1 - x_2\|} : x^* \in S_{X^*}, x_1 \neq x_2 \in X, \frac{\langle x^*, x_1 - x_2 \rangle}{\|x_1 - x_2\|} = 1 \right\},$$

and its *numerical radius* by

$$w_L(F) = \sup \{ |\lambda| : \lambda \in W_L(F) \}.$$

The *Lipschitz numerical index* of the Banach space X is then defined by

$$\begin{aligned} n_L(X) &= \inf \{ w_L(F) : F \in \text{Lip}_0(X), \|F\|_L = 1 \} \\ &= \max \{ \lambda \geq 0 : \lambda \|T\|_L \leq w_L(T) \text{ for every } T \in \text{Lip}_0(X) \}. \end{aligned}$$

Notice that the canonical inclusion $\mathcal{L}(X) \hookrightarrow \text{Lip}_0(X)$ is a linear isometry, and that if F is linear $W_L(F) = W(F)$ and $w_L(F) = w(F)$. Therefore,

$$n_L(X) \leq n(X).$$

A central question posed in [20] is whether equality always holds. The answer to this question depends on whether the underlying field is $\mathbb{R}$ or $\mathbb{C}$.

Figure 1: The paper defines the Lipschitz numerical index and records the central equality question.

index of $(\mathbb{C}^2, \|\cdot\|_\infty)$ is one (by the aforementioned result of Kadets et al.), while the Lipschitz numerical index of $(\mathbb{C}^2, \|\cdot\|_2)$ is zero, as shown above. It is therefore natural to ask which values are realized by the pair $(n(X), n_L(X))$ as X ranges over all complex Banach spaces. However, we do not pursue this question in the present work.

Real case. In contrast to the complex case, no known example shows that $n_L(X)$ and $n(X)$ may be different when X is a real space. This motivates the conjecture $n_L(X) = n(X)$ for every real Banach space X . In fact, partial results support this claim: Wang et al. proved that if X has the Radon-Nikodým property (RNP) then the equality holds; moreover, they showed that if X is a real lush space, then both indices coincide and are equal to one. In particular, if X does not contain a copy of ℓ_1 or X is *strongly regular*,

Figure 2: In the real case the source states the conjectural equality $n_L(X) = n(X)$ for every real Banach space.

In this paper, we contribute further evidence to this conjecture. We prove that the equality $n(X) = n_L(X)$ holds if X is a (real) separable Banach space (Section 2) or if X is a (real) dual Banach space (Section 3). The arguments are based on two distinct techniques for constructing linear operators from Lipschitz maps that preserve certain properties: differentiation via convolution with a Gaussian probability measure, and the use of invariant means.

2. Real separable Banach spaces

We first state the main result of the section.

Figure 3: The source paper proves the equality for real separable Banach spaces. It also proves it for real dual Banach spaces.

Definitions and external inputs

For a Banach space X , write $\text{Lip}_0(X)$ for the Banach space of Lipschitz maps $F : X \rightarrow X$ with $F(0) = 0$, equipped with the Lipschitz norm

$$\|F\|_L = \sup_{x \neq y} \frac{\|F(x) - F(y)\|}{\|x - y\|}.$$

The Lipschitz numerical radius is denoted $w_L(F)$, and

$$n_L(X) = \inf\{w_L(F) : F \in \text{Lip}_0(X), \|F\|_L = 1\}.$$

For linear operators, w_L agrees with the classical numerical radius w , hence always $n_L(X) \leq n(X)$.

We use the following published inputs.

1. Perez-Hernandez proves that $n_L(Y) = n(Y)$ for every real separable Banach space Y , and also for every real dual Banach space Y .
2. Wang–Huang–Tan prove that if

$$X = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{c_0}, \quad X = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{\ell_1}, \quad \text{or} \quad X = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{\ell_\infty},$$

then

$$n_L(X) = \inf_{\gamma \in \Gamma} n_L(X_\gamma).$$

3. The corresponding classical numerical-index formula is

$$n(X) = \inf_{\gamma \in \Gamma} n(X_\gamma)$$

for the same three sums. This formula is classical; it is also recorded in the absolute-sum literature of Martin–Meri–Popov–Randrianantoanina.

Claim

Theorem 1. *Let $\{X_\gamma : \gamma \in \Gamma\}$ be a family of real Banach spaces such that*

$$n_L(X_\gamma) = n(X_\gamma) \quad (\gamma \in \Gamma).$$

Let X be one of the three sums

$$\left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{c_0}, \quad \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{\ell_1}, \quad \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{\ell_\infty}.$$

Then

$$n_L(X) = n(X).$$

Proof. By the Wang–Huang–Tan stability theorem for the Lipschitz numerical index,

$$n_L(X) = \inf_{\gamma \in \Gamma} n_L(X_\gamma).$$

By the hypothesis on the summands this is

$$\inf_{\gamma \in \Gamma} n(X_\gamma).$$

Finally, the classical stability formula for the numerical index of c_0 -, ℓ_1 -, and ℓ_∞ -sums gives

$$\inf_{\gamma \in \Gamma} n(X_\gamma) = n(X).$$

Therefore $n_L(X) = n(X)$. □

Corollary 1. *If each X_γ is either a real separable Banach space or a real dual Banach space, then the three sums above satisfy $n_L(X) = n(X)$.*

Proof. Perez-Hernandez proves $n_L(Y) = n(Y)$ for every real separable Banach space and for every real dual Banach space. Apply the theorem to the family $\{X_\gamma\}$. □

Relation to previous run work

The earlier packet

`solutions/partial/2507.18208.c0_sum_lipschitz_numerical_index/`

proved the special case of c_0 -sums of separable real spaces by a direct countable-coordinate reduction. The present packet gives a broader but more literature-driven partial result: it also covers ℓ_1 - and ℓ_∞ -sums and any summand class for which the equality is known, including real dual spaces.

A route not promoted

A more ambitious separable-projection reduction was tested. Suppose every separable subset of a real Banach space X is contained in a separable range E of a norm-one projection $P : X \rightarrow E$. Given a norm-one Lipschitz map $F : X \rightarrow X$, one can choose E containing a near Lipschitz-norm-attaining pair x, y and the two values $F(x), F(y)$. Then the compressed map $G = PF|_E : E \rightarrow E$ satisfies $\|G\|_L$ almost 1 and

$$w_L(G) \leq w_L(F).$$

Since E is separable, Perez-Hernandez gives

$$w_L(F) \geq w_L(G) \geq n(E)\|G\|_L.$$

However, this does not imply $w_L(F) \geq n(X)$ unless one also knows the classical comparison $n(E) \geq n(X)$. That comparison is available for the coordinate summands in the three absolute sums used above, but it is not a consequence of arbitrary one-complementability alone. For that reason, the projectional-skeleton route is recorded only as an attempt, not as a theorem.

Literature and novelty check

Searches were made for the source arXiv id, the phrases *Lipschitz numerical index*, *projectional skeleton*, *one-complemented*, and *separable complementation*, together with the exact equality $n_L(X) = n(X)$. The relevant hits were the source paper, Wang–Huang–Tan’s 2014 stability paper, Choi–Jung–Tag’s 2025 paper on one-complemented increasing unions, and the classical absolute-sum paper of Martin–Meri–Popov–Randrianantoanina. No later paper was found explicitly stating the corollary above as a partial answer to the 2025 real-case question.

Verification notes

This packet is a theorem-by-combination. The proof is complete if the two stability formulas are accepted as external inputs. The human verifier should focus on:

- confirming that Wang–Huang–Tan’s formula applies to arbitrary families under the c_0 , ℓ_1 , and ℓ_∞ norms;
- confirming the matching classical numerical-index formula for the same three sums;
- checking that the packet is understood as a partial result and not a solution for arbitrary real Banach spaces.

References

- [1] A. Perez-Hernandez, *Lipschitz vs Linear Numerical Index in certain Banach spaces*, arXiv:2507.18208.
- [2] R. Wang, X. Huang, and D. Tan, *On the numerical radius of Lipschitz operators in Banach spaces*, J. Math. Anal. Appl. 411 (2014), 1–18; arXiv:1211.5753.
- [3] M. Martin, J. Meri, M. Popov, and B. Randrianantoanina, *Numerical index of absolute sums of Banach spaces*, J. Math. Anal. Appl. 375 (2011), 207–222; arXiv:1003.3269.
- [4] G. Choi, M. Jung, and H.-J. Tag, *On the Lipschitz numerical index of Banach spaces*, Collect. Math. 76 (2025), 81–103; arXiv:2110.13821.